AMAN CHANDAK

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SKILLS

Motion Planning & Control: Trajectory Optimization, Configuration Spaces, Path Planning, LQR, Model Predictive Control (MPC), Sliding Mode Control, Rigid Body Dynamics, Sim-to-Real Transfer
State Estimation & Autonomy: SLAM, 3D Pose Estimation, Probabilistic Filtering (EKF/UKF), Sensor Fusion, Computer Vision, VIO, 3D Data Processing
Languages & Tools: C++, Python, ROS 2, MATLAB, Simulink, Linux, Git, Docker, MuJoCo, PyTorch

EXPERIENCE

Student Research Aide — Feb 2025 – Present
RISE Lab, Arizona State University (Honeywell-funded project) — Mesa, AZ

- Architected an end-to-end 5G Positioning Reference Signal (PRS) localization testbed for GNSS-denied indoor UAVs, integrating synchronized Ettus USRP software-defined radios and OpenAirInterface.
- Engineered a detector-aware probabilistic residual multilayer perceptron model to predict per-link multipath bias and uncertainty.
- Fused learned corrections and variances into a weighted Gauss-Newton observed time difference of arrival (OTDOA) solver, reducing held-out test horizontal RMSE from 11.827 m to 2.080 m.
- Developed a hardware-matched MATLAB ray-tracing simulation pipeline generating a training corpus of 2,157,161 valid links to model leading-edge detector dynamics.

Sr. UAV Engineer — Jan 2021 – Jun 2024
Aerial IQ India Pvt Ltd — Gurugram, India

- Led the development of perception-assisted navigation systems for high-altitude UAVs, integrating inertial sensors with radar-based velocity sensing (HRVS) for robust flight in visually degraded environments.
- Engineered and deployed the nation's first commercial tethered drone system, managing hardware-software integration for continuous surveillance applications.
- Developed automated control loops in Python/MAVLink for the Automatic Tether Station (ATS), ensuring system stability through rigorous field testing and sensor calibration.
- Validated perception and control stacks in real-world scenarios, debugging sensor noise and actuation latency to ensure safety-critical reliability.

PROJECTS

AI-Powered Mobile Manipulation with Stretch Robot (NICE Lab, ASU)

- Developing a semantic scene understanding pipeline integrating Vision-Language-Action Models (VLAs) to parse natural language instructions into spatial navigation goals.
- Utilizing transformers for multi-modal reasoning, enabling the robot to perceive objects in 3D space and execute manipulation tasks.
- Implementing the full autonomy stack in ROS 2, bridging high-level AI reasoning with low-level perception and actuation.

Non-Prehensile Robotic Manipulation Optimization – EEE587
Formulated a mixed-integer optimization for contact forces in MuJoCo and benchmarked LQR versus MPC for hybrid dynamical systems.

Lateral Stability via Advanced ADAS Control – MAE506
Designed LQR and Sliding Mode Controllers in MATLAB and Simulink for vehicle stability using state-space modeling.

EDUCATION

Arizona State University, Tempe, AZ — Aug 2024 – May 2026
M.S. Robotics and Autonomous Systems — GPA: 4.0/4.0
University of Petroleum and Energy Studies, Dehradun, India — Aug 2016 – May 2020
B.Tech Aerospace Engineering (Avionics)

PUBLICATION

"5G-based Localization for Unmanned Aerial Vehicles" – **AIAA SciTech Forum 2026**